Algebra (All rings are commutative with 1)

concepts and propsitions 1. Rings and Ideals

1.1.1 Every ring $A \neq 0$ has at least one maximal ideal

pf (sketched): Let Σ be the set of all ideals $\neq (1)$ in A.

then Σ is a non-empty, partially ordered set. ~~then by Zorn's Lemma,~~

let a_α be a chain of ideals in Σ. where $\forall \alpha, \beta \in I$., $a_\alpha \subseteq a_\beta$ or $a_\beta \subseteq a_\alpha$

let $a = \bigcup_\alpha a_\alpha$ then a is an ideal $(1 \notin a \in \Sigma) \Rightarrow$ by Zorn's Lemma, Σ has a maximal element. ▨

Prop. I.1.2. ① The set $\mathfrak{N}$ of all nilpotent elements in a ring A is an ideal, and $A/\mathfrak{N}$ has no nilpotent elements $\neq 0$. called nilradical of A.

② The nilradical of $A = \bigcap_{\mathfrak{p} \in A} \mathfrak{p}$. , $\mathfrak{p}$ is the prime ideal in A.

pf (skipped).

Def I.1.3. The Jacobson radical $\mathfrak{R}$ of $A := \bigcap_{m \in A} m$. m is the maximal ideal in A

Prop. I.1.4. $x \in \mathfrak{R} \iff 1 - xy$ is a unit in A

Lemma I.1.5. i). Let A be a ring, $m \neq (1)$ s.t. $\forall x \in A - m$ is a unit in A.

Then A is a local ring and m its maximal ideal.

ii). m a maximal ideal of A. s.t. $\forall x \in 1 + m$ is a unit. then A is a local ring.

pf of I.1.4. (sketched). by using I.1.5 ▨

Def I.1.6. (Operations on ideals). ① $a+b$: the set of all $x+y$ where $x \in a$. $y \in b$.

② $\sum_{i \in I} a_i$: the set of sums $\sum_{i \in I} x_i$ all but finitely are zeros.

③. ab : the ideal generated by all xy where $x \in a$. $y \in b$.

④ (ideal quotient) $(a:b) = \{x \in A : xb \subseteq a\}$. is an ideal.

In particular, note $(0:b) = \mathrm{Ann}(b)$.

⑤. (radical) the radical of a is $r(a) = \{x \in A : x^n \in a \text{ for some } n > 0\}$. is an ideal.

Prop. I.1.7. Define a ring homomorphism $\phi : A \longrightarrow \prod_{i=1}^n (A/a_i)$.

i). If a_i, a_j are coprime $(\forall i \neq j)$, then $\prod a_i = \bigcap a_i$

ii). ϕ is surjective $\iff$ a_i, a_j coprime $(\forall i \neq j)$ (Chinese remainder theorem.)

iii). ϕ is injective $\iff$ $\bigcap a_i = (0)$.

Prop. I.1.8. i) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be prime ideals and let a be an ideal $\subseteq \bigcup_{i=1}^n \mathfrak{p}_i$ then $a \subseteq \mathfrak{p}_i$ for some i.

ii) Let $\mathfrak{p} \supseteq \bigcap_{i=1}^n a_i$ then $\mathfrak{p} \supseteq a_i$ for some i.

pf of 1.8. (find it in the last lecture note).

Lemma I 1.9 (prime avoidance) A ring. I, $P_1 \dots P_r$ ideals, P_i ($i \geq 3$) are prime

If $\forall i \in I$ we have $I \not\subseteq P_i$, then $I \not\subseteq \bigcup_{i=1}^{r} P_i$

pf (skipped). by induction on r. ▨

Prop. I.1.10. $\sqrt{a}$ is the intersection of the prime ideals which contain a.

pf: Apply 1.2 on A/a ▨

Prop. I.1.11. Let D be the set of zero-divisors of A. then $D = \bigcup_{x \neq 0} r(\mathrm{Ann}(x))$.

pf: $D = r(D)$ ▨

Prop. I.1.12. If $r(a)$ & $r(b)$ are coprime. then a and b are coprime.

Def. I.1.13. ~~Extension~~: Let $f: A \to B$ be a ring homomorphism. $a \subseteq A$, $b \subseteq B$ ideal.

① the extension of a: a^e = the ideal generated by $f(a)$.

② the contraction of b: $b^c = f^{-1}(b)$ the ideal.

Prop. I.1.14. i). $a \subseteq a^{ec}$, $b \supseteq b^{ce}$.

ii) $b^c = b^{cec}$; $a^e = a^{ece}$.

pf: (skipped).

2. Modules

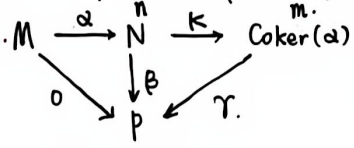
Def I.2.1. $f: M \to N$ an A-module homomorphism, then $\mathrm{Coker}(f) = N/\mathrm{Im}(f)$

Def I.2.2. N, P be submodules of R-module M. $(N:P) := \{r \in R : rP \subseteq N\}$.

$\mathrm{Ann}(M) := (0:M)$.

Remark: fix homomorphism: $\begin{cases} \kappa: N \to \mathrm{Coker}(\alpha) \\ \beta: N \to P. \end{cases}$ $\beta\alpha = 0$. then $\exists!\ \gamma: \mathrm{Coker}(\alpha) \to P$.

(universal property).



$\left(\begin{array}{l} \beta\alpha = 0. \\ \forall m \in \mathrm{Coker}(\alpha) \cdot \exists n \in N \text{ s.t. } m = k(n) \Leftrightarrow \gamma(m) = \beta(n) \end{array}\right)$

Def I.2.3. $\dots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \dots$ is said to be exact iff $\mathrm{Im}(f_i) = \mathrm{Ker}(f_{i+1})$

(1)

Prop. I.2.4. (1) ~~as~~ can be divided as: some short exact sequences:

$N_i = \mathrm{Im}(f_i) = \mathrm{Ker}(f_{i+1})$. then $0 \to N_i \to M_i \to N_{i+1} \to 0$ is exact.

Prop I.2.5. $\forall \lambda \in \Lambda$. TFAE:

(1) $\forall \lambda \in \Lambda$. $M'_\lambda \to M_\lambda \to M''_\lambda$ exact

(2) $\bigoplus M'_\lambda \to \bigoplus M_\lambda \to \bigoplus M''_\lambda$ exact

(3) $\prod M'_\lambda \to \prod M_\lambda \to \prod M''_\lambda$ exact

...ke lemma) Consider the two exact rows

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \to & 0 \\ \downarrow\gamma & & \downarrow\tau & & \downarrow\gamma'' & & \\ 0 \to N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' & & \end{array}$$

then there exists an exact sequence:

$$\ker(\gamma') \xrightarrow{\varphi} \ker(\tau) \xrightarrow{\psi} \ker(\gamma'') \xrightarrow{\partial} \operatorname{Coker}(\gamma') \xrightarrow{\psi'} \operatorname{Coker}(\tau) \xrightarrow{\psi''} \operatorname{Coker}(\gamma'')$$

Corollary I.2.7. (9 lemma).

$$\begin{array}{ccccccccc} & & 0 & & 0 & & & & \\ & & \downarrow & & \downarrow & & & & \\ 0 & \to & L' & \to & L & \to & L'' & \to 0 & (5) \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & M' & \to & M & \to & M'' & \to 0 & (4) \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & N' & \to & N & \to & N'' & \to 0 & (6) \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \\ & & (1) & & (2) & & (3) & & \end{array}$$

(1), (2), (3), (4) are exact.

then (5) exact $\Leftrightarrow$ (6) exact.

Thm. I.2.8. (i) $M' \to M \to M'' \to 0$ exact $\Rightarrow$ $\forall$ module N, $0 \to \operatorname{Hom}(M'', N) \to \operatorname{Hom}(M, N) \to \operatorname{Hom}(M', N)$ exact.

(ii). $0 \to N' \to N \to N''$ exact $\Rightarrow$ $\forall$ module M, $0 \to \operatorname{Hom}(M, N') \to \operatorname{Hom}(M, N) \to \operatorname{Hom}(M, N'')$ exact.

II. Tensor Product.

Def. II.1.1. M, N, P are R-modules. then $\exists$ R-module T and R-bilinear $g: M \times N \to T$ has the Prop. property:

① $\forall$ R-module P. and R-bilinear $f: M \times N \to P$, $\exists! f': T \to P$. s.t. $f = f' \circ g$.

② If (T, g), (T', g') are the solution of ump, then $\exists!$ j isomorphism, $T \to T'$ s.t. $j \circ g = g'$. then we will have $\operatorname{Bil}_R(M, N; P) \cong \operatorname{Hom}(M \otimes N; P)$.

And we can define $T = M \otimes N$.

$$\begin{array}{ccc} M \times N & \xrightarrow{g} & T := M \otimes N \\ & \searrow^{f} & \downarrow \exists! f' \text{ s.t. } f = f' \circ g \\ & & P \end{array}$$

Prop. II.1.2. If $x_i \in M$, $y_i \in N$, $\sum x_i \otimes y_i = 0$ in $M \otimes N$. Then there exists $M' \subseteq M$. $N' \subseteq N$ finitely generated. s.t. $\sum x_i \otimes y_i = 0$ in $M' \otimes N'$.

Prop. II.1.3. M, N, P are R-modules, then there exists isomorphism:

(i) $M \otimes N \cong N \otimes M$. (ii) $(M \otimes N) \otimes P \cong M \otimes N \otimes P \cong M \otimes (N \otimes P)$.

(iii). $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$. (iv). $R \otimes M \cong M$.